

Auditing Emergent Physics Claims in Rewriting Systems: Exact Tests of Dimension, Causal Invariance, and the Second Law

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The Wolfram Physics Project claims that simple rewriting systems exhibit emergent spatial dimension, relativistic structure from causal invariance of the rewriting, and a second law of thermodynamics from computational irreducibility. Each of these claims rests on an inference pattern that the present campaign has already made exactly measurable for its own purposes, so the claims can be audited rather than debated. This paper reports six audits, run under the campaign's sealed-instrument discipline and framed so that a defender of the program would accept every pass condition as fair. A shell-based dimension estimator is exact on controls of known dimension and refuses the diverging control, while the naive ball-volume slope that casual dimension measurements use reads low on every exact control by a finite-radius bias that small-graph dimension claims inherit. A manifestly reversible cellular automaton, built on the featured rule of the program's own second-law writings, raises its declared coarse-grained entropy from an ordered seed and then retraces the curve bitwise exactly with exact recovery of the initial microstate, so the entropy growth is observational, which sharpens the program's observer-relative account of the second law into a measured statement. For string rewriting, a causally invariant sorting rule agrees exactly under every update order, a planted non-confluent rule is flagged with three distinct terminal strings, and a growth rule splits into conserved counts that are invariant and an arrangement that belongs to the update order. Causal-graph invariants built from token genealogy are identical across update orders where causal invariance holds, and extremal foliations achieve the causal depth bound exactly while random foliations spend up to twice as many slices, so simultaneity bookkeeping moves while the invariants stay frozen. A hypergraph rule's measured dimension then passes a held-out scaling test, beating a preregistered exponential alternative forty-fold in held-out error. The capstone audit shows that the same rule's bulk bookkeeping is not invariant. Five declared match orders give different relation counts, different vertex counts, and a seventeen percent spread in the dimension estimate, so the surviving dimension claim carries a measured prescription caveat at these sizes. The verdict pattern is uniform. Quantities are exact where an algebraic property protects them, prescription-borne where matching conflicts exist, and the second law of reversible computation is observational entropy. Nothing in this paper bears on whether the program is correct as fundamental physics.

I. INTRODUCTION

The Wolfram program proposes that fundamental physics emerges from simple rewriting systems. Space and its dimension are read from the growth of neighborhood volumes in an evolving hypergraph [1], special relativity is attributed to causal invariance of the rewriting, the property that every order of applying the rules yields the same causal structure [2], quantum mechanics is read from the branching of a multiway evolution [3], and the second law of thermodynamics is attributed to computational irreducibility acting on computationally bounded observers [5]. The program's claims have been debated mostly at the level of plausibility, because the inference patterns they rest on, reading a dimension from a growth curve, asserting independence from an update order, counting structures relative to a slicing, and locating an entropy in an observer, are rarely stated with pass and failure conditions that both sides accept in advance.

This campaign built exact instruments for all four of those inference patterns before turning to the Wolfram

program, for reasons internal to its own questions. A foliation-covariance audit asks whether a claimed event survives a passive change of the observer's slicing, and the campaign's fold theorem states exactly which changes of multiplicity are legitimate under such a change. A prescription audit asks whether a claimed quantity survives a change of update prescription, after an earlier stage of the campaign proved that a published threshold belonged to a discretization rather than to the dynamics. An estimator-validation rule forbids any instrument from measuring anything until it has passed exact controls, including a control it must refuse. And the campaign's entropy track separated observational entropy from thermodynamic entropy by exhibiting a reversible system whose observed entropy swings while its hidden ensemble is recovered exactly [7, 8].

These audits test claims, not people, every pass condition is one a defender of the program would accept as fair, every failure condition is a finding about a model class rather than a criticism of a research effort, and a claim that survives comes out stronger than it went in.

The paper reports six audits. An instrument net validates the dimension estimator on exact controls and, along the way, measures the finite-radius bias of the naive estimator that casual dimension claims use. A reversible

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cellular automaton makes the program’s second-law account a measured statement. A string-rewriting audit measures which quantities survive a change of update order, with a planted non-confluent rule as the trap the pipeline must catch. A causal-graph audit checks the same question on causal structure and adds a foliation sweep. A hypergraph audit tests a dimension claim on held-out data it was never fitted to. The capstone audit then turns the update-order question on the hypergraph rule itself and finds its bulk bookkeeping prescription-borne. A closing section assembles the scorecard, which claims are invariant, which are observational, and which belong to the prescription.

All systems are finite and computable, all measurements are exact or carry declared tolerances, and no result bears on whether the Wolfram program is true or false as fundamental physics. A claim shown prescription-borne here is a result about a declared rule at measured sizes under a declared audit, nothing more.

II. THE CLAIMS AND THE AUDIT INSTRUMENTS

The track document of this campaign records a mapping with one row per program claim, the instrument that audits it, and what a pass and a failure mean. Four rows were run and are reported here.

Emergent dimension from ball-volume growth is audited by the estimator net of Sec. III followed by the held-out scaling test of Sec. VII. A pass means the rule’s estimated dimension is finite, stable on radii never used for fitting, and beats preregistered alternative growth laws. A failure means the dimension estimate belongs to the fitted radius window, a statement about estimator scope on that rule.

Causal invariance, the basis of the program’s relativistic claims, is audited by the prescription protocols of Secs. V and VIII, with the update order as the prescription. For a rule with provable confluence the declared invariants must agree exactly across independent update orders. For a rule without such a proof the order dependence is measured rather than assumed away, and any quantity that varies is classified prescription-borne for that rule.

Structures counted relative to a slicing are audited by the foliation protocol of Sec. VI. The campaign’s fold theorem allows an honest middle outcome, a multiplicity that varies across slicings while a declared signed invariant is conserved, so the audit never demands naive invariance of raw counts.

The second law from computational irreducibility is audited by the reversible-automaton protocol of Sec. IV. This row is not adversarial at all. The program’s own account already locates the second law in the observer [5], and the audit supplies the exact-arithmetic witness that turns that account from an argument into a measurement.

The fifth row of the mapping, quantum mechanics from multiway branching [3], remains design only and no experiment for it is reported here.

TABLE I. The estimator net on exact controls. The shell estimator of Eq. (1) is compared with the naive ball-volume slope on the same graphs, windows, and base points. The tree control must be refused, and is, with the recorded number a non-refusing estimator would print shown in parentheses. All numbers are from the committed evidence record.

Control	Exact d	Shell $\hat{d}$	Naive slope	Deficit
path	1	1.000	0.896	0.104
2D torus	2	2.000	1.771	0.229
3D torus	3	2.958	2.657	0.343
regular tree	none	refused (6.47)	5.490	none

Two standing disciplines bind every run. Every pipeline carries a planted artifact, a deliberately non-confluent rule whose order dependence is known in advance, and a pipeline that fails to flag its own trap voids the run. And every window, coarse-graining, prescription list, and alternative model is declared before the run it governs, per the track’s anti-circularity contract.

III. THE DIMENSION-ESTIMATOR NET AND THE NAIVE-SLOPE BIAS

No dimension is quoted in this paper before the instrument that measures it has passed exact controls, including a control it must refuse. For a graph, let $S(v, r)$ be the number of vertices at graph distance exactly r from a base vertex v , and let $B(v, r)$ be the number within distance r . The campaign’s estimator fits the slope of $\log S$ against $\log r$ over the window $2 \leq r \leq 10$, declared before any rule measurement, averages over 24 declared base points, and reports

$$\hat{d} = 1 + \frac{d \log S}{d \log r}, \quad (1)$$

with a sealed tolerance of 0.15 and a refusal condition. If the local log-log slope grows across the window instead of stabilizing, the graph is flagged non-finite-dimensional and no dimension is quoted.

Table I shows the net. The path graph returns 1.000 with local slopes flat to machine precision, the two-dimensional torus returns 2.000 within 2×10^{-16} , and the three-dimensional torus returns 2.958, a deviation of 0.042 against the tolerance of 0.15, with local slopes climbing from 1.919 toward 2 across the window as the lattice shells converge on their asymptotic law. The regular tree, the control the estimator must refuse, shows local slopes growing from 2.71 to 10.43 across the window with no stabilization, so the recorded refusal condition fires and the graph is flagged non-finite-dimensional. The record retains the number a non-refusing estimator would print, 6.47, and quoting it as a dimension would fail the net as badly as misreading a torus, because the point of the net is that the instrument must know what it cannot measure.

The net also produced a finding about the estimator the program’s informal practice suggests, the log-log slope of the ball volume itself. On the same exact controls that slope reads 0.896, 1.771, and 2.657 against exact values of 1, 2, and 3. The origin is elementary. A ball volume growing as r^d with a subleading term of relative order $1/r$ has a log-log slope that sits below d by a term of order $1/r$ at every finite radius, and over practical windows the accumulated deficit is 0.10 to 0.34 in the measured cases. Any dimension read from a ball-volume slope on a small graph inherits this bias, and it acts in one direction, toward reading low. The shell form of Eq. (1) removes the accumulation by differentiating first, which is why it is the instrument used in Secs. VII and VIII.

IV. THE SECOND LAW OF A REVERSIBLE AUTOMATON IS OBSERVATIONAL

The program’s second-law claim is already observer-relative. Computational irreducibility makes the microdynamics opaque to a bounded observer, and the entropy that observer assigns grows [5]. The campaign’s entropy track proved a sharpened form of this statement for reversible hidden dynamics, an observed binned entropy that swings while the hidden ensemble is recovered exactly, so that the entire change is observational [7, 8]. This audit transplants that protocol onto the automaton family the program itself uses.

The system is a second-order reversible elementary automaton,

$$a_{t+1} = F(a_t) \oplus a_{t-1}, \quad (2)$$

where F is an elementary rule applied to the current row and $\oplus$ is addition modulo two. The update is invertible by construction, since $a_{t-1} = F(a_t) \oplus a_{t+1}$ recovers the past exactly, in the family reviewed by Toffoli and Margolus [6] and explored in the program’s foundational book [4]. The declared instance takes F to be Rule 122, the featured example of the program’s own second-law writings [5], on 1024 cells with periodic boundary for 1500 steps, from a random 256-cell center region with the prior row zero. The declared coarse-graining is the Shannon entropy of the empirical histogram of non-overlapping four-cell blocks, declared before the run. The arithmetic is exact over the two-element field, so the reversed trajectory is not approximately the forward trajectory, it is the forward trajectory bitwise, or the implementation is wrong.

The measured result. The coarse-grained entropy rises from 1.646 bits at the ordered seed to 3.288 bits at the turnaround, a rise of 1.641 bits against a declared bar of 1.0. Under reversal the entropy curve retraces with a maximum defect of exactly 0.0 bits at every mirrored time, and the initial pair of configurations is recovered exactly, cell for cell. The entropy of the finite reversible system fluctuates along the way, as it must, and the claim is the rise from the ordered seed together with the exact retrace, not monotonicity.

The reading is the campaign’s entropy-track conclusion verbatim. The dynamics loses nothing, the observer’s coarse-graining produces the entire entropy change, and the second law of this reversible computational system is observational entropy. This is the program’s own account [5], converted into a statement with an exact witness, and any second-law claim for reversible rewriting systems now has a measured form to be compared against.

Three instrument findings bound the protocol’s scope. First, the reversible form of Rule 30, the program’s flagship pseudorandomizer, runs from a single seed into a dense phase in which the rule’s output vanishes on saturated neighborhoods and the block entropy falls, so that rule and seed pair never randomizes under the declared coarse-graining. Second, the symmetric initialization that sets the prior row equal to the current row makes the orbit time-reflection symmetric, and Rule 122 then shows a near-recurrence toward order within a thousand steps, so the initialization must be asymmetric for the rise to be sustained. Third, eight-cell byte blocks are too coarse for structured automaton fields and four-cell blocks resolve the growth. A fourth note concerns the bar itself. An early version of the reversal swapped the wrong pair of rows, a one-step misalignment, and the exact-retrace bar caught it immediately, which is what a bar set at exactly zero is for.

V. STRINGS UNDER A CHANGE OF UPDATE ORDER

Causal invariance is the program’s stated basis for relativistic claims [2]. Its operational content is that the physics of a rewriting system should not depend on the order in which applicable rules are applied. The campaign audits exactly that, with the update order as a prescription in the sense of its earlier threshold finding, under three declared orders, leftmost match, rightmost match, and seeded random match, all from the same declared initial string of 60 letters over the alphabet $\{A, B\}$.

The causally invariant control is the sorting rule $BA \rightarrow AB$. Each event removes exactly one inversion, one misordered pair of letters, so the algebra fixes the event count in advance. The measured result agrees exactly. Under every one of the three orders the rule terminates in the same sorted string after exactly 406 events, and 406 is the inversion number of the declared initial string. The invariance is not approximate and not statistical. It is the confluence of the rule made visible, and it is what a genuine causal-invariance claim looks like when measured.

The planted trap is the rule pair $\{AB \rightarrow B, BA \rightarrow A\}$, built to be non-confluent because overlapping matches conflict. The pipeline must flag it or the run is void. It fires. The three orders terminate in three distinct strings, B , $BBBB$, and BB , after 7, 4, and 6 events respectively. Everything about this rule’s outcome, its terminal state, its event count, its history, belongs to the update order, and the pipeline records it as prescription-borne. The

trap discharges the campaign’s standing obligation to prove that the pipeline can catch the artifact class it is hunting.

The growth rule $A \rightarrow AB$, run for 200 declared events, splits along exactly the line the campaign’s fold theorem predicts. The conserved quantities agree exactly across all three orders, two A letters and 201 B letters in a final string of length 203, because each event preserves the A count and adds exactly one B regardless of where it applies. The arrangement of those letters differs between orders. Counts protected by a conservation law are prescription-invariant while the configuration carrying them is prescription-borne, the signed-versus-unsigned split of the campaign’s earlier work measured here in rewriting form.

VI. CAUSAL GRAPHS AND FOLIATIONS

The previous section audited terminal states. This one audits the causal structure itself, the object the program’s relativistic claims live on [2]. The causal graph is built from token genealogy. Each event consumes tokens and produces tokens, and a causal edge runs from the event that produced a token to the event that consumed it. The audit runs the sorting rule on a declared 24-letter string under leftmost and rightmost orders and compares declared invariants of the two causal graphs, then sweeps foliations of one graph.

For the sorting rule the causal-graph invariants are identical across prescriptions. Both orders give 55 events, 88 causal edges, causal depth 13, and the same out-degree multiset, one event of out-degree zero, twenty of out-degree one, and thirty-four of out-degree two. The causal structure of the confluent rule is a prescription-independent object, measured rather than asserted.

The trap’s causal graphs differ radically, as they must. The leftmost order gives 7 events joined by 6 causal edges, while the rightmost order gives 4 events with no causal edges at all, a causally disconnected history in which every event consumes only initial tokens. For a non-confluent rule not even the causal graph’s gross shape survives the prescription, which is the precise sense in which relativistic claims require causal invariance rather than merely invoke it.

The foliation sweep then measures the observer’s remaining freedom on the one invariant causal graph. A valid foliation is a sequence of antichains respecting the causal partial order and exhausting the graph. The earliest and latest layerings both use exactly 13 slices, achieving the causal depth bound exactly, with maximum slice width 7. Five seeded random foliations use 19 to 26 slices with maximum widths of 4 and 5. The invariants are frozen while the number of slices an observer spends, the simultaneity bookkeeping, moves by a factor of two between the extremal and the sampled random foliations. This is the campaign’s foliation lesson in rewriting form. What varies under a change of slicing is the observer’s

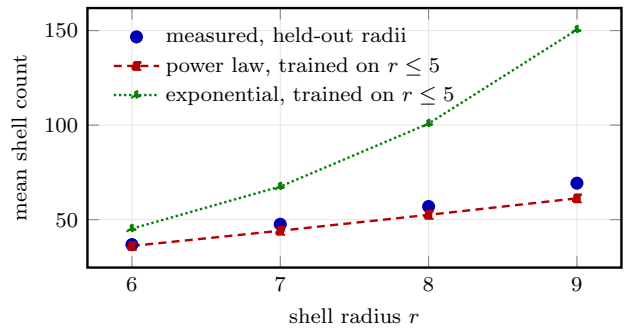


FIG. 1. The held-out test of the hypergraph dimension claim. Mean shell counts of the rule of Eq. (3) on the held-out radii, never used in fitting, against the predictions of the two preregistered growth laws calibrated on the training radii alone. The power law tracks the measured shells while the exponential alternative overshoots increasingly with radius. Every point is read from the committed evidence record.

accounting, and what survives is the partial order, so any claimed structure must be checked against the invariants and not against the accounting.

VII. A DIMENSION CLAIM THAT SURVIVES HELD-OUT SCALING

With the estimator net passed, the audit turns to the program’s dimension claim on a hypergraph rule of the program’s own class [1]. The declared rule is

$$\{\{x, y\}, \{x, z\}\} \rightarrow \{\{x, z\}, \{x, w\}, \{y, w\}, \{z, w\}\}, \quad (3)$$

where w is a fresh vertex, run from a double self-loop for 11 generations under one declared prescription, generational maximal scan-ordered non-overlapping matching. The final hypergraph has 1794 relations on 897 vertices. The shell estimator of Sec. III is calibrated on the training radii $2 \leq r \leq 5$ over 30 declared base points, giving $\hat{d} = 2.305$, and the claim is then tested on the held-out radii $6 \leq r \leq 9$, which were never used in fitting. Two preregistered growth laws, the finite-dimension power law and the exponential alternative that a non-finite-dimensional graph would follow, are each calibrated on the training window and made to predict the held-out shell counts.

Figure 1 shows the held-out comparison. The power law predicts the four held-out shell counts with a summed squared error of 0.028 in log space, while the exponential alternative accumulates 1.09, a factor of forty in favor of the finite-dimension reading. The measured shells at radii 6 through 9 are 36.8, 47.6, 57.0, and 69.3, and the power-law predictions track them to within eight counts. This dimension claim genuinely survives the audit at the measured sizes. The measurement is at finite radius on a finite object, a crossover beyond the held-out window cannot be excluded, and the asymptotic dimension claim

TABLE II. The five prescriptions of the capstone audit on the growth rule of Eq. (3). Every column that a naive reading would treat as physics of the rule differs across the declared match orders. All numbers are from the committed evidence record.

Prescription	Relations	Vertices	Degree sum	$\hat{d}$
storage order	1794	897	3588	2.454
reversed	1882	941	3762	2.883
seeded shuffle a	1710	855	3420	2.664
seeded shuffle b	1810	905	3620	2.431
by second vertex	1758	879	3510	2.751

of the program is not settled by any finite-size audit. A second caveat is measured in the next section.

VIII. THE CAPSTONE, HYPERGRAPH BOOKKEEPING IS PRESCRIPTION-BORNE

The string audit of Sec. V found exact prescription invariance where confluence protects it. The capstone audit asks the same question of the hypergraph rule of Eq. (3), whose left-hand side consumes two relations sharing a vertex, so that candidate matches overlap and a maximal non-overlapping matching depends on the order in which candidates are scanned. Five match orders were declared in advance, the storage order of the relation list, that order reversed, two seeded random permutations, and a sort by second vertex. Each order evolves the same initial condition for the same 11 generations, and the same declared quantities are measured on each result over 24 base points to radius 9, a configuration distinct from the WM-4 calibration, which fitted the training radii with thirty base points, so the storage order reads 2.454 here where WM-4 read 2.305, and the prescription caveat is measured in this capstone configuration.

Table II and Fig. 2 show the result. The five prescriptions give five different evolutions. Relation counts span 1710 to 1882, vertex counts span 855 to 941, degree sums span 3420 to 3762, and the shell-estimator dimension spans 2.43 to 2.88, a seventeen percent spread against the audit's five percent invariance bar. Because matches conflict, even the number of events per generation is a prescription choice, so the bulk bookkeeping of this rule, its sizes, its degree structure, and its measured dimension, is prescription-borne at these sizes. The contrast with the string audit is the finding. The sorting rule was exact under every order because an algebraic property, confluence with a conserved inversion count, protected it. The growth rule has no such protection, and the audit measures the consequence.

The WM-4 verdict of Sec. VII therefore inherits a measured caveat. Its dimension of 2.305 was measured under the storage-order prescription, and the same rule under other declared orders yields dimensions up to 2.88. One part of the verdict is robust. The power law beats the ex-

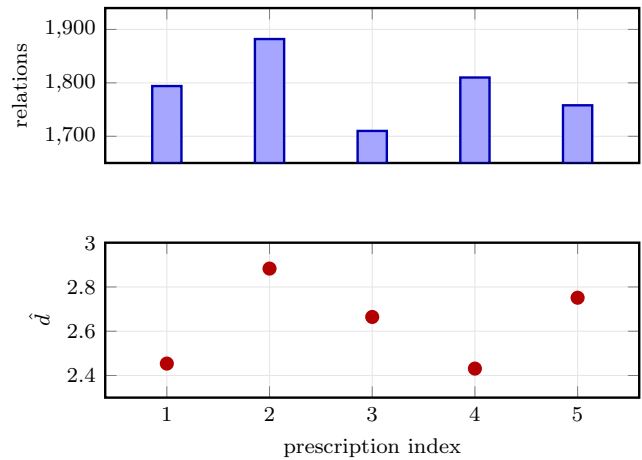


FIG. 2. The capstone audit, one growth rule under five declared match orders. Prescription indices are 1 the storage order, 2 the reversed order, 3 and 4 two seeded random permutations, and 5 the sort by second vertex. The upper panel shows the final relation count and the lower panel the shell-estimator dimension of the same evolution. Both move with the prescription, spanning 1710 to 1882 relations and 2.43 to 2.88 in dimension. Every value is read from the committed evidence record.

ponential alternative under every one of the five prescriptions, with error ratios from six-fold to several hundred-fold, so the qualitative finding of finite-dimensional scaling at measured sizes survives the prescription sweep, while the quantitative value of the dimension does not. A dimension claim for a rule with match conflicts must either declare its prescription or report a prescription band, and at these sizes that band is wide.

The planted trap of this audit is a contraction rule on a twelve-relation path with a three-branch fork at its midpoint, run to exhaustion. It fires as required, terminating in three distinct final states across the prescriptions, and the pipeline flags it. The trap's own construction preserved a lesson. The bare contraction path without the fork was found confluent en route, any contraction order composes the path to a single edge, so a trap must contain a genuine conflict and not merely an aesthetic of disorder. The fork supplies the conflict, and the recorded lesson is part of the audit's evidence.

IX. THE SCORECARD

Assembled across the six audits, the verdicts sort into three classes, and the sorting principle is visible in every row.

Invariant, exactly. The sorting rule's terminal state and event count, protected by confluence and equal to the inversion number. The conserved letter counts of the growth rule, protected by a conservation law. The causal-graph invariants of the confluent rule, identical across prescriptions. The causal depth bound, achieved

exactly by the extremal foliations. And, robustly though not exactly, the qualitative finite-dimension verdict of the hypergraph rule, which survives both a held-out test and a five-prescription sweep.

Observational. The second law of the reversible automaton. The entropy rise of 1.641 bits is real for the declared observer and the dynamics underneath it loses nothing, with retrace defect exactly zero and exact microstate recovery. This class is where the program’s own second-law account already lives, now with a measured witness.

Prescription-borne. The arrangement of the growth rule’s letters. The entire outcome of every planted non-confluent rule. And the bulk bookkeeping of the flagship hypergraph rule, its relation counts, its vertex counts, its degree structure, and its quantitative dimension at these sizes, spread across a seventeen percent band by declared match orders.

The pattern is uniform and is the paper’s summary finding. A quantity is exact across observers and prescriptions precisely where an algebraic property protects it, a confluence proof, a conservation law, an antichain bound, and it is prescription-borne wherever matching conflicts exist and no such protection applies. Claims of the Wolfram program that rest on protected quantities measured with validated instruments survive these audits and come out stronger, as the fair-framing clause intended. Claims that rest on bulk bookkeeping of rules with match conflicts inherit a measured prescription caveat, and the naive dimension estimator adds a one-directional bias on top. None of this decides the program’s physics. It decides which of the program’s numbers are about the rules and which are about the bookkeeping.

X. WHAT THIS PAPER ESTABLISHES

Table III registers the claims with their labels. Read together they audit four of the program’s five claimed inference patterns. Dimension claims survive where the

instrument is validated, the test is held out, and the prescription band is declared. Causal invariance is exact where confluence protects it and is measurably absent where it does not. Structure counting is legitimate on the invariants of the causal partial order and illegitimate on the slicing’s bookkeeping. The second law of reversible computation is observational entropy. The multiway quantum claim remains unaudited and no statement about it is made.

XI. METHODS AND REPRODUCIBILITY

All audits follow the campaign’s standing discipline. Estimator windows, tolerances, coarse-grainings, prescription lists, base-point counts, seeds, and alternative growth laws were declared before the runs they govern, per the track’s anti-circularity contract, and the fair-framing clause of the track document, that the audits test claims rather than people and that a surviving claim comes out stronger, governs the reading of every verdict. Every result-bearing run writes an append-only JSON evidence record containing the complete declared parameter set, outcome arrays, and a canonical SHA-256 hash of the record body. The records for this paper are `wm0-dimension.json`, `wm1-reversible-ca.json`, `wm2-prescription.json`, `wm3-foliation.json`, `wm4-hypergraph.json`, and `wm2h-prescription.json` in the campaign repository, each carrying the exploratory label of the campaign’s evidence-grading scheme, with the committed runners alongside them. Every prescription audit carries a planted non-confluent trap whose detection is a validity condition of the run, and both traps fired. Every number and every figure coordinate in this paper is read from those committed records, and the figure tables are generated deterministically by a committed script with no free parameters. A reducer may summarize records but may not rewrite them.

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TABLE III. Claim register for the Wolfram-models track. Measured in model, exploratory label, means an exact measurement in a declared construction with append-only evidence records, declared prescriptions and windows, and planted traps where the audit class requires them, whose evidence record carries the exploratory label because no seal governs the run, pending the cross-substrate replication clause. The label below applies to each measurement inside its declared scope.

Result	Claim	Label
Estimator net	the shell dimension estimator is exact on controls of known dimension and refuses the diverging tree control	measured in model, exploratory label
Naive-slope bias	the log-log ball-volume slope reads low by 0.10 to 0.34 on exact controls at practical radii, a bias any small-graph dimension claim inherits	measured in model, exploratory label
Reversible second law	the declared coarse entropy of Rule 122 in second-order reversible form rises 1.646 to 3.288 bits and retraces bitwise exactly with exact microstate recovery, so the growth is observational entropy	measured in model, exploratory label
String prescription audit	the confluent sorting rule agrees exactly under every declared update order with event count equal to the inversion number, the planted non-confluent rule is flagged with three distinct terminals, and the growth rule splits into invariant conserved counts and prescription-borne arrangement	measured in model, exploratory label
Causal-graph audit	causal-graph invariants from token genealogy are identical across prescriptions for the confluent rule, the trap's causal graphs differ radically, and extremal foliations achieve the depth bound exactly while random foliations spend up to twice the slices	measured in model, exploratory label
Held-out dimension	the declared hypergraph rule's finite-dimension power law beats the preregistered exponential alternative forty-fold in held-out error at measured sizes	measured in model, exploratory label
Capstone prescription audit	five declared match orders give the growth rule different relation counts, vertex counts, and dimension estimates spanning a seventeen percent band, so its bulk bookkeeping is prescription-borne and the held-out dimension claim carries a measured prescription caveat at these sizes	measured in model, exploratory label
Scope	nothing in this paper bears on whether the Wolfram program is true or false as fundamental physics, and no result may be quoted as evidence in either direction	non-claim